Rules for Assigning an Oxidation Number (O.N.)

General Rules

1. For an atom in its elemental form (Na, C, O_2, Cl_2, P_4, etc.): O.N. = 0.

2. For a monatomic ion: O.N. = ion charge (with the sign before the numeral; e.g. Mg^{2+} has an O.N. of +2).

3. The sum of O.N. values for the atoms in a molecule or formula unit of a compound equals zero. The sum of O.N. values for the atoms in a polyatomic ion equals the ion's charge. (e.g. Carbonate CO_3^{2-} $\quad ON_{\mathbf{Carbon}} + 3\,ON_{\mathbf{Oxygen}} = -2$)

For Group 1:	O.N. = +1 in all compounds.
For Group 2:	O.N. = +2 in all compounds.
For Hydrogen:	O.N. = +1 in combination with nonmetals.
For Fluorine:	O.N. = –1 in all compounds.
For Oxygen:	O.N. = –1 in peroxides (e.g., H_2O_2);
	O.N. = –2 in all other compounds (except with F).
For Group 17 (other halogens):	O.N. = –1 in combination with metals, nonmetals (except O), and halogens lower in the group.

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